



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix



Executive Summary

- Methodologies

- Data Collection utilizing SpaceX API and Web Scraping
- Data Wrangling and Manipulation
- Exploratory Data Analysis using SQL and Visualization
- Interactive Visual Analysis utilizing Folium and Plotly Dash
- Classification modeling and Predictive Machine Learning

- Results

- Exploratory Data Analysis Insights
- Interactive Dashboard for Visual Analytics
- Predictive Analytics via Machine Learning model



Introduction

- The space industry is heating up, with SpaceX leading the way through their reusable rockets. SpaceX can recover the first stage of some of their launches, which allows them to offer commercial rocket launches for \$62 million, a great price when compared to other providers of the same service which can cost up to \$165 million per launch.
- SpaceY is an emerging competitor in the commercial rocket launch market and seeks to determine the price of each launch. Using information gathered about SpaceX, our goal is to build an interactive dashboard for our stakeholders modeled after SpaceX's launch data as well as build a machine learning model to predict if SpaceX will reuse the stage on a launch.
- This will aid SpaceY in determining the factors that influence if a rocket will land successfully, the success rate that is achievable, and the ideal operating conditions to ensure a successful landing, ultimately allowing them to compete with SpaceX.

Section 1

Methodology

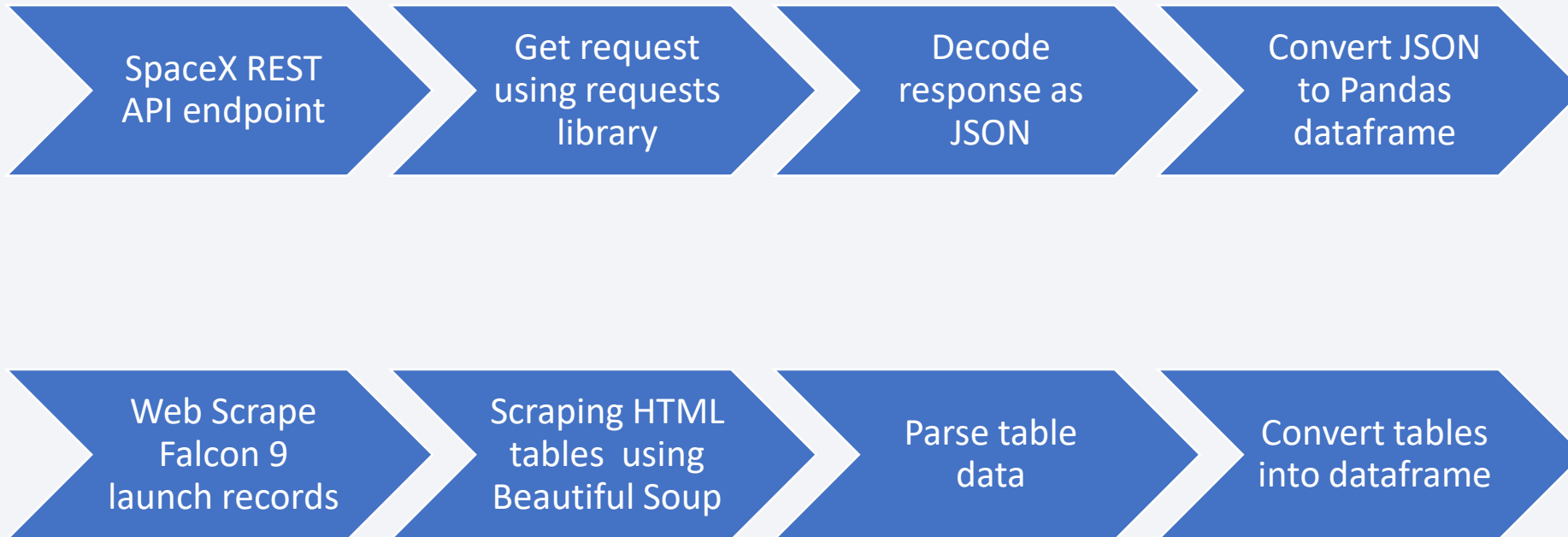
Methodology

Executive Summary

- Data collection methodology:
 - Data was gathered via SpaceX REST API and web scraping of Wikipedia launch data
- Perform data wrangling
 - Conversion of categorical features utilizing one-hot encoding to prepare them for our learning models
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Building, tuning, and evaluation of various classification models to build the most effective machine learning model

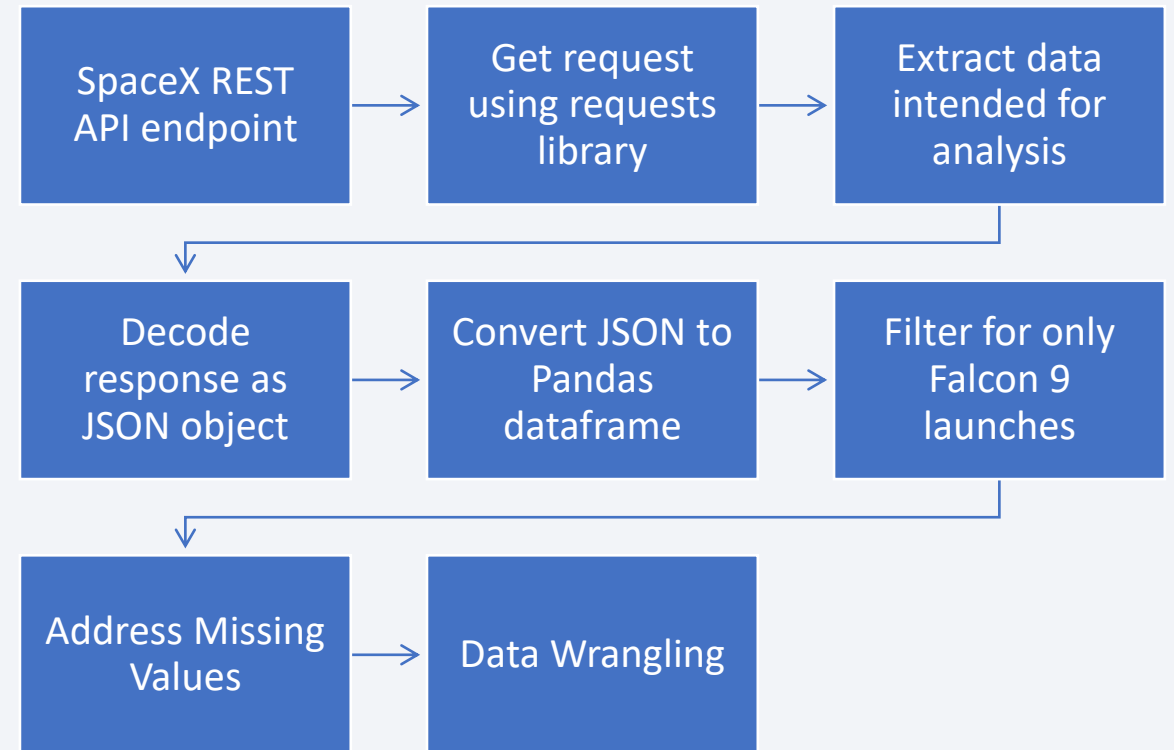
Data Collection

- All data was gathered via SpaceX REST API requests and web scraping of Wikipedia



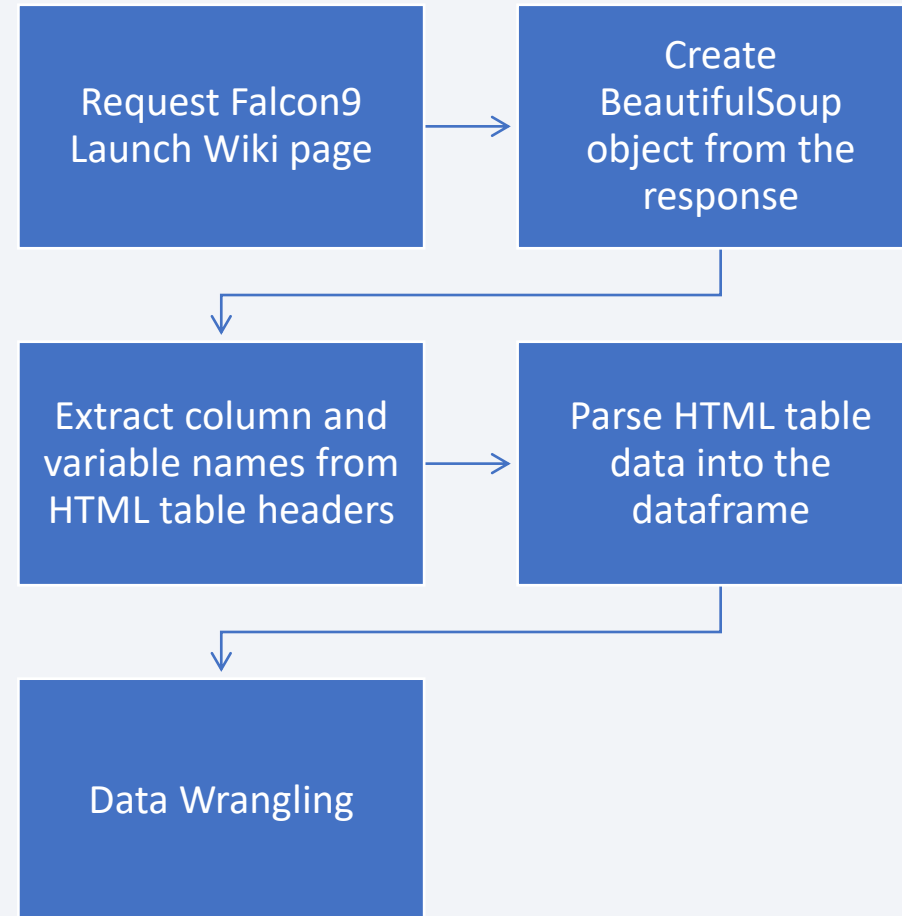
Data Collection – SpaceX API

- Using a get request on SpaceX's API, we extract the data we are interested in (booster name, launch site, payload mass, landing site, etc.) as a JSON object before converting to a dataframe to conduct cleaning and wrangling.



Data Collection - Scraping

- Using BeautifulSoup within Python allowed for web scraping of Falcon 9 launch records on Wikipedia.
- After requesting data from the URL and extracting the table we are interested in from the HTML tables, the data can be converted into a dataframe for wrangling and analysis.



Data Wrangling

- Missing values will be addressed first, followed by determining the data types of each column
- Data was processed for Exploratory Data Analysis to find trends and determine the training labels for our model.



EDA with Data Visualization

- Scatterplots were utilized for the following visuals:
 - Relationship between Flight # and Payload
 - Relationship between Flight # and Launch Site
 - Relationship between Payload and Launch Site
 - Relationship between Flight # and Orbit type
 - Relationship between Payload and Orbit type
- Other Visualization types:
 - A bar chart was utilized to visualize the success rate of each orbit type.
 - A line chart was utilized to show yearly trends of launch success

Build an Interactive Map with Folium

- Map objects created for the Folium map:
- Folium.Circle: added to highlight a specific coordinate on the map with a circle
- Folium.Marker: added to highlight a specific coordinate on the map with a marker
- MarkerCluster: simplifies a map that contains multiple markers in proximity or with the same coordinates
- Mouseposition: allows user to view coordinates of a point on the map when hovered over with the cursor
- Folium.polyline: an object used to draw a line between a launch site and its closest city, railway, and highway

Build a Dashboard with Plotly Dash

- The final dashboard allows the user to utilize a drop-down menu to filter by different launch sites
- Utilizing callback functions, when the user selects an option in the dropdown, it updates the charts within the visualization
- A Pie chart shows the ratio of successes and failures at each site
- A scatter plot shows the relationship between Outcome and Payload Mass for the various booster versions
- A range slider was added to filter by Payload

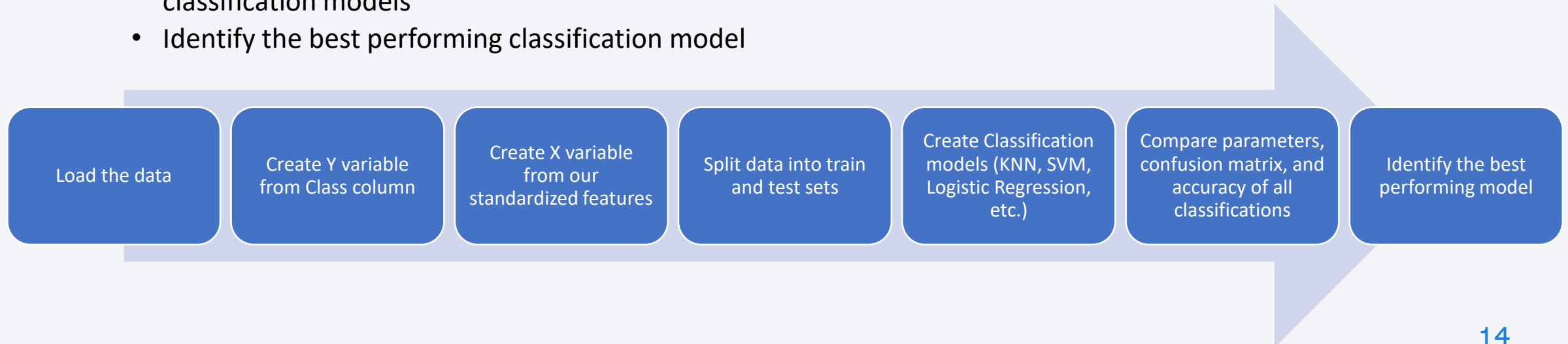
Link to the notebook hosted on Github:

https://github.com/jtnickel/Python/blob/main/IBM%20Capstone%20-%20Dash_app.py



Predictive Analysis (Classification)

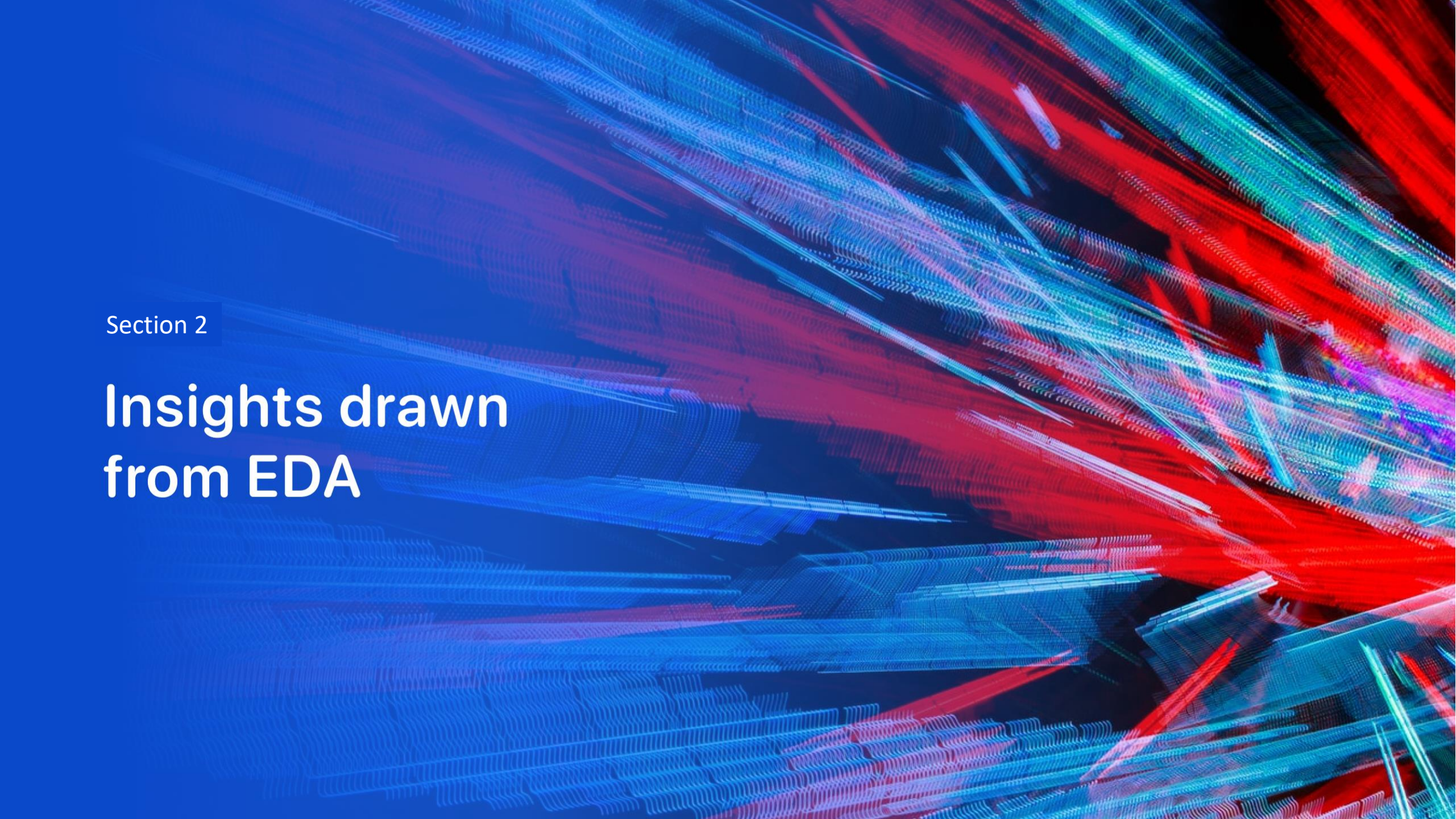
- The following outlines the process of developing and testing our models to predict whether the first stage will land or not:
 - Utilized Numpy to create an array from the Class column
 - Standardization of data
 - Split our data into train and test sets
 - Calculating the most ideal hyperparameters for KNN, SVM, Decision tree and Logistic Regression classification models
 - Identify the best performing classification model



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



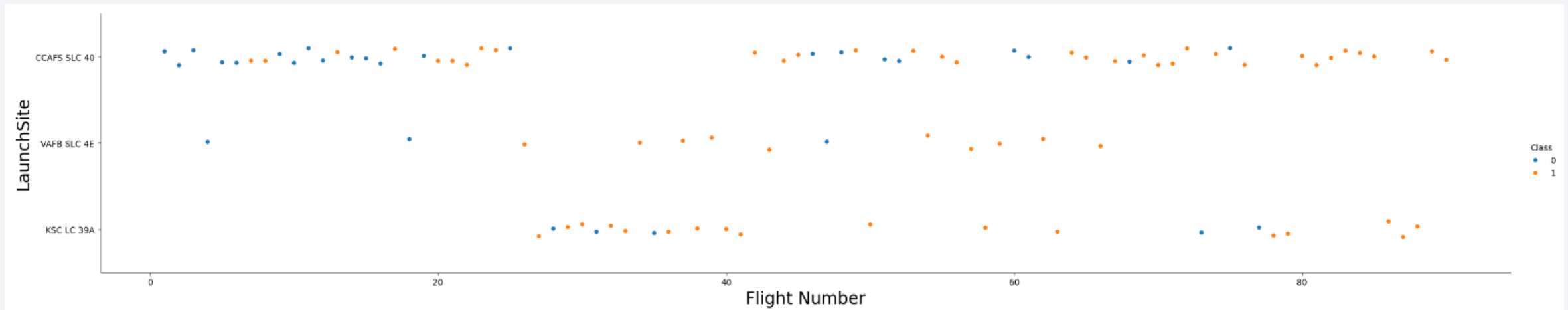
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

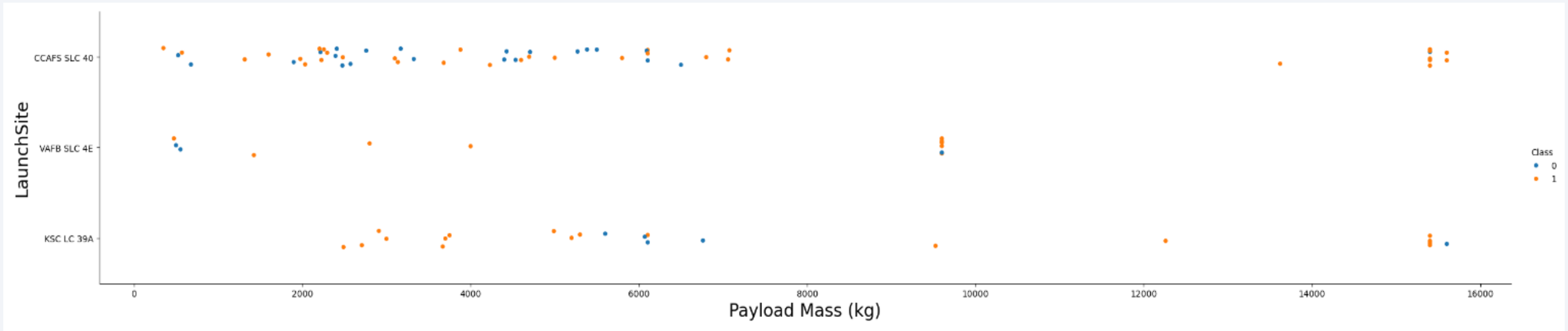
Insights drawn from EDA

Flight Number vs. Launch Site

- This scatter plot shows there is a higher success rate among Launch Sites with a greater number of flight launches



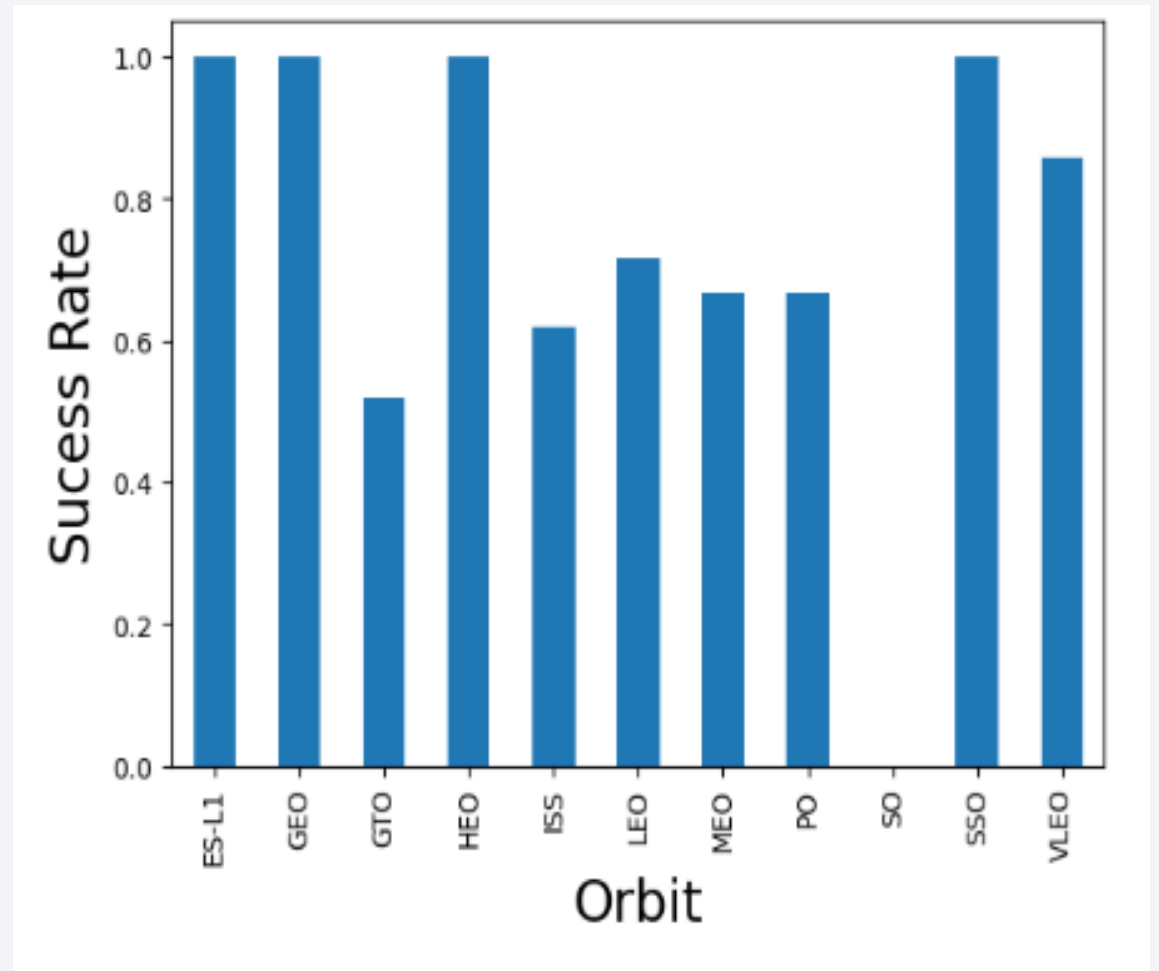
Payload vs. Launch Site



- Notably from this plot, we see that CCAFS SLC 40 has a very high success rate of flights with very heavy payloads
- At Launch Site VAFBSLC 40, no launches have been at or above 10,000 kg

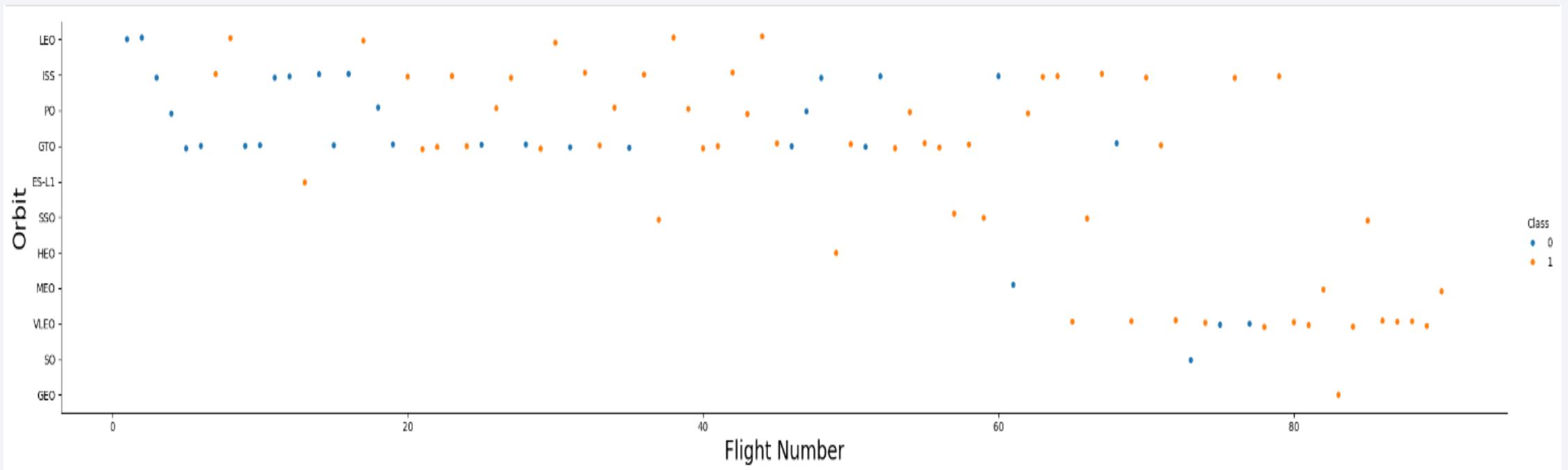
Success Rate vs. Orbit Type

- We can see the most successful orbit patterns are:
 - ES-L1
 - GEO
 - HEO
 - SSO



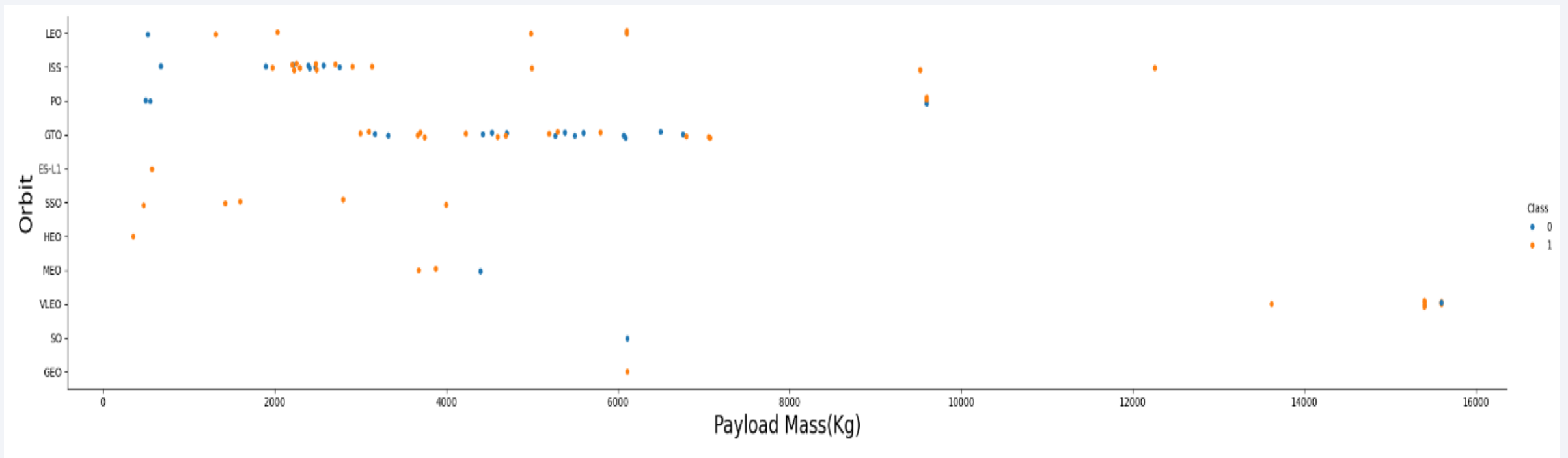
Flight Number vs. Orbit Type

- Some orbit types have a high success rate over multiple flights (LEO, VLEO) whereas other orbit types high success rate comes alongside a small sample (SSO, HEO, GEO)



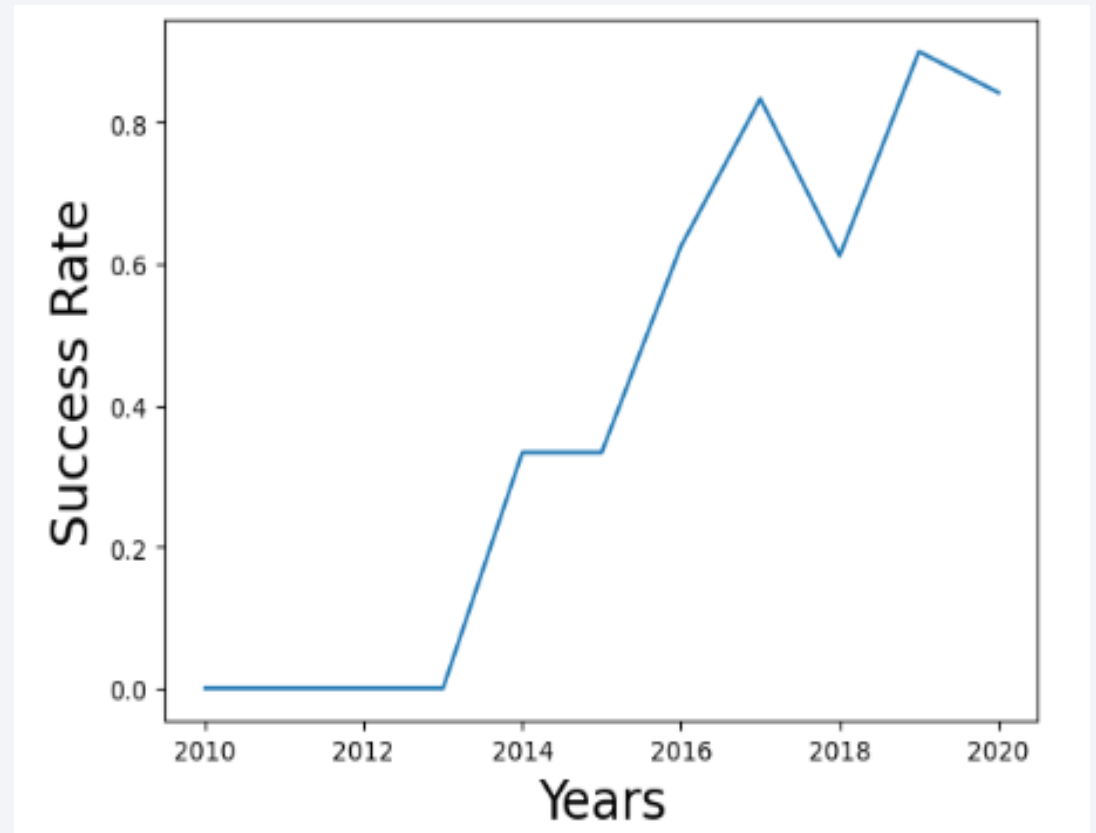
Payload vs. Orbit Type

- For heavier payloads, VLEO, ISS and PO orbits all achieve very high success rates
- Inversely, those same 3 orbit types have the lowest success rate at lighter payloads ($> 2000\text{kg}$)



Launch Success Yearly Trend

- From 2013 to 2017 there is rapid and mostly steady increase in success rates
- Success rate drops approximately 20% during 2017 but recovers the following year

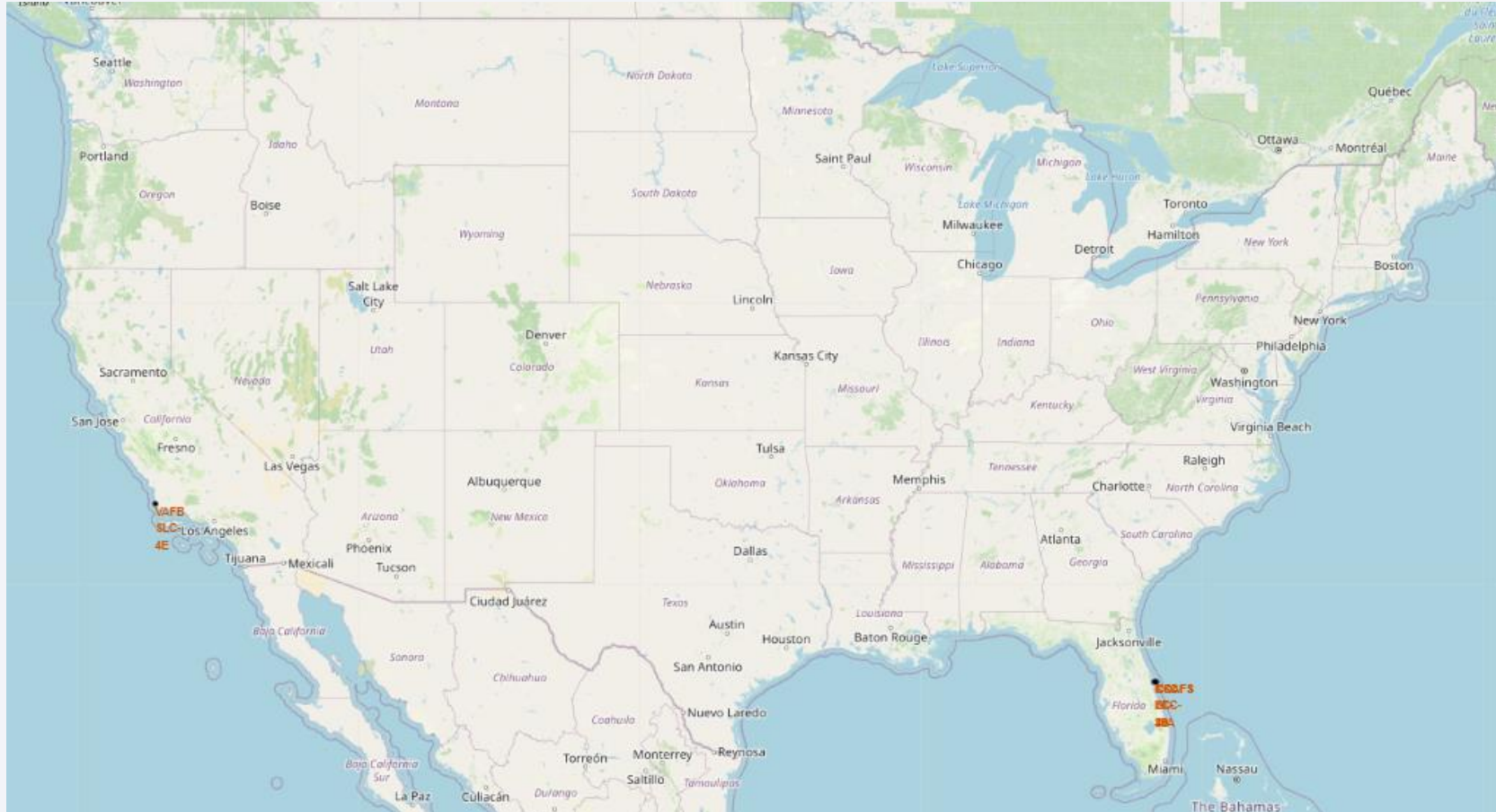


A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

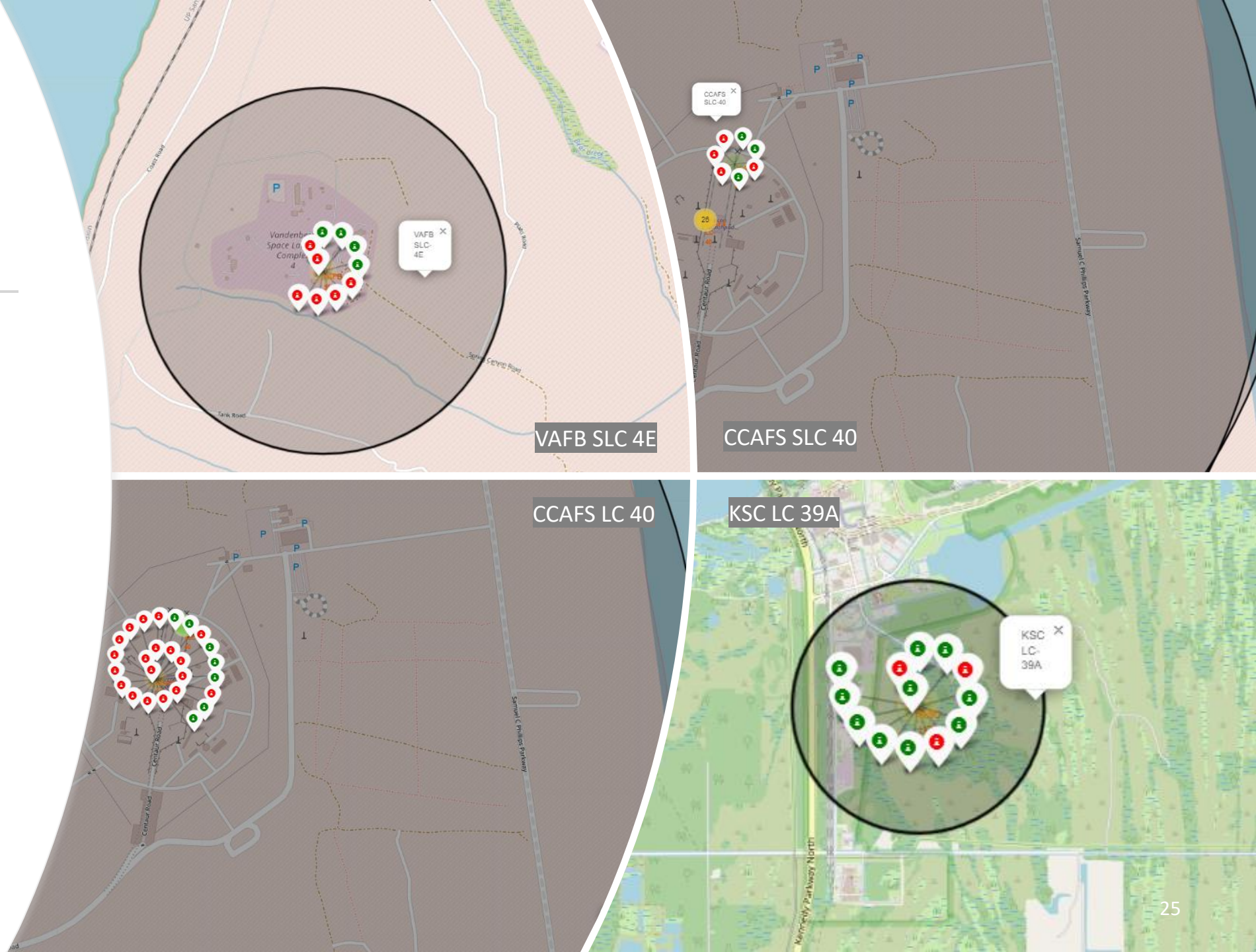
Launch Site Locations



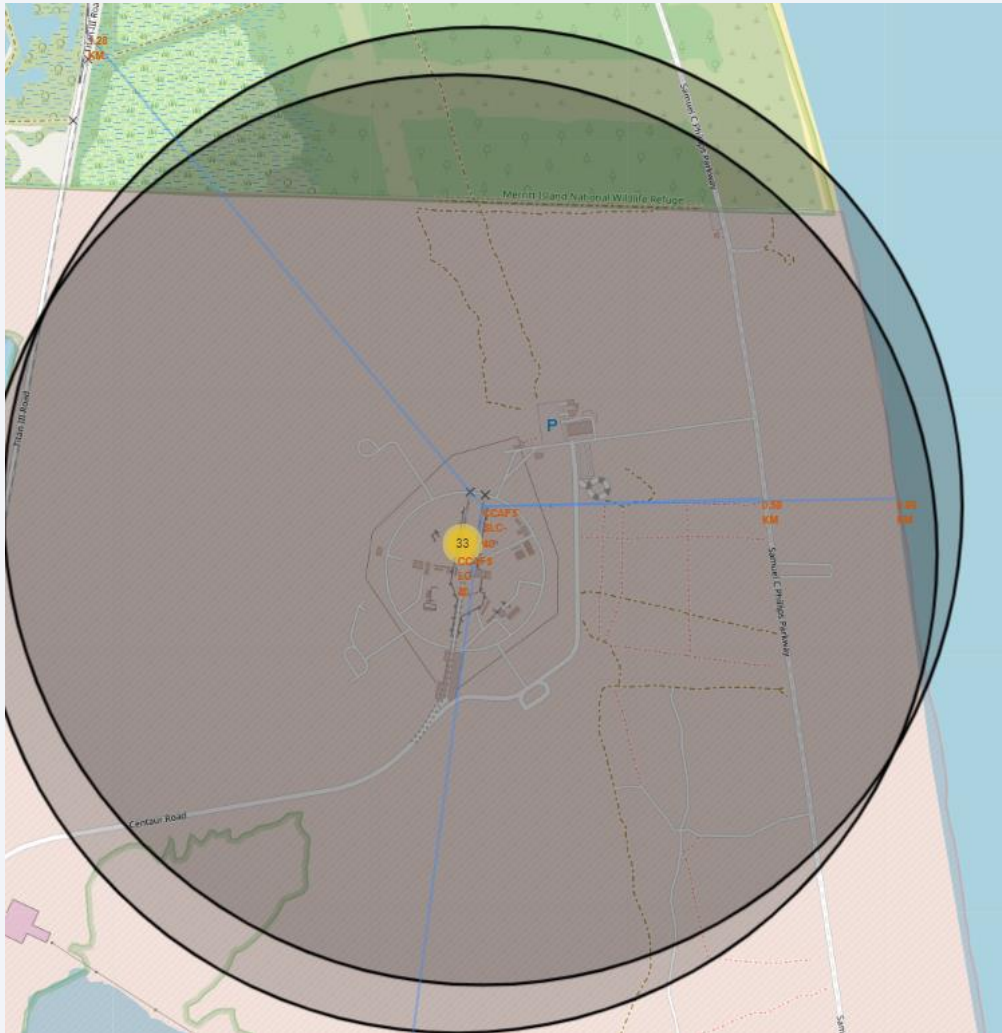
- 3 of the test sites are in Florida, 1 is in California
- All sites are in relatively close proximity to the ocean and the equator

Successes and Failures by Launch Site

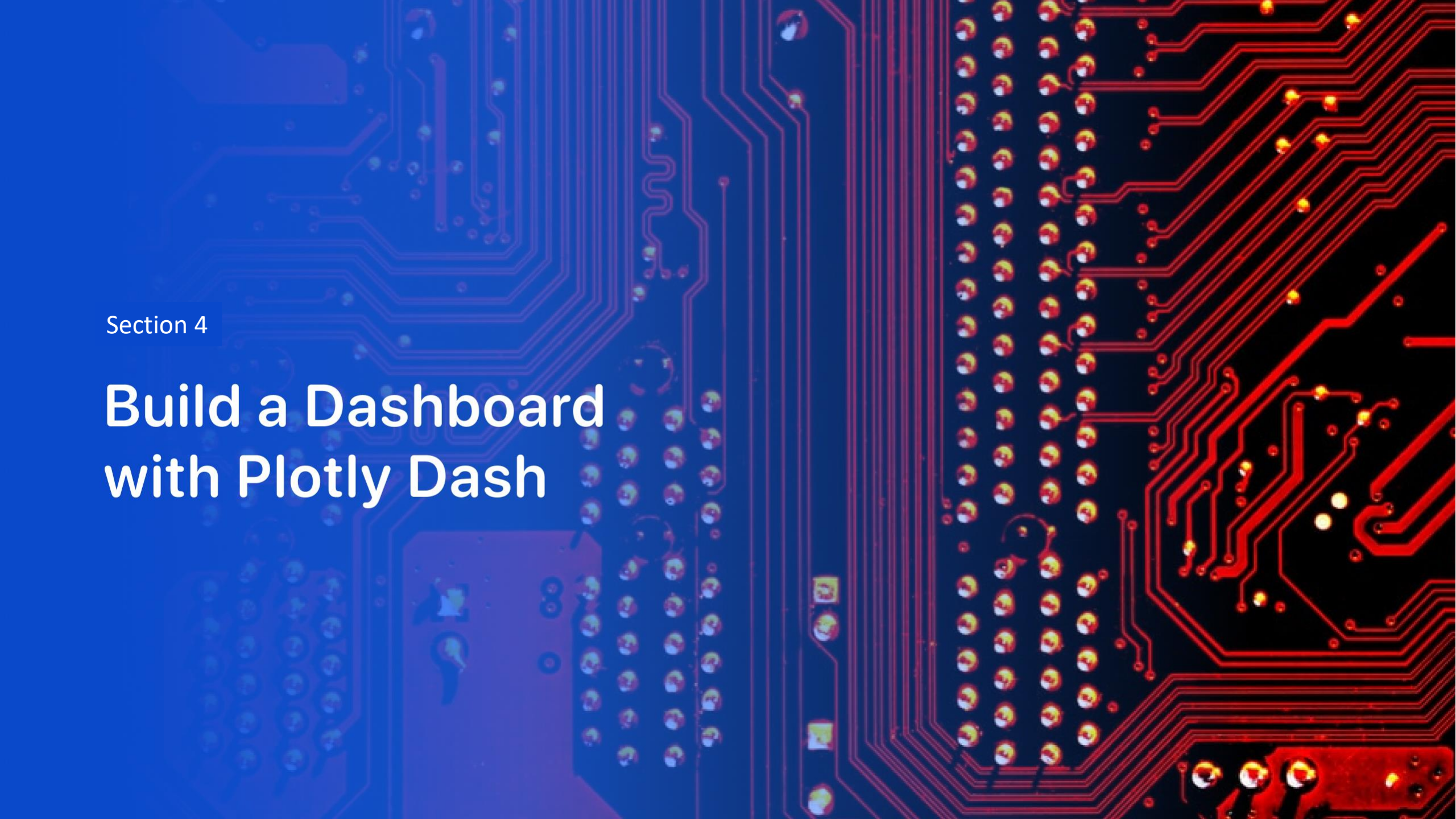
- Green markers show a successful launch, whereas red markers indicate a failed launch
- The California launch site can be seen in the bottom right image
- CCAFS SLC 40 and CCAFS LC 40 are in very close proximity to one another



What's near a Launch Site? What isn't?



- The thin blue lines on the right indicate the distance to the nearest coastline (.86km), the nearest highway (.58km), and the nearest railway (1.28km) for the two furthest east launch sites
- Notably, all 4 launch sites are very near at least one highway, one railway and are all on the coastline
- The highways and railways are ideal for bringing in heavy cargo and for workers to get to work
- All 4 launch sites are far from populous heavy areas and close to the ocean which are both likely safety precautions

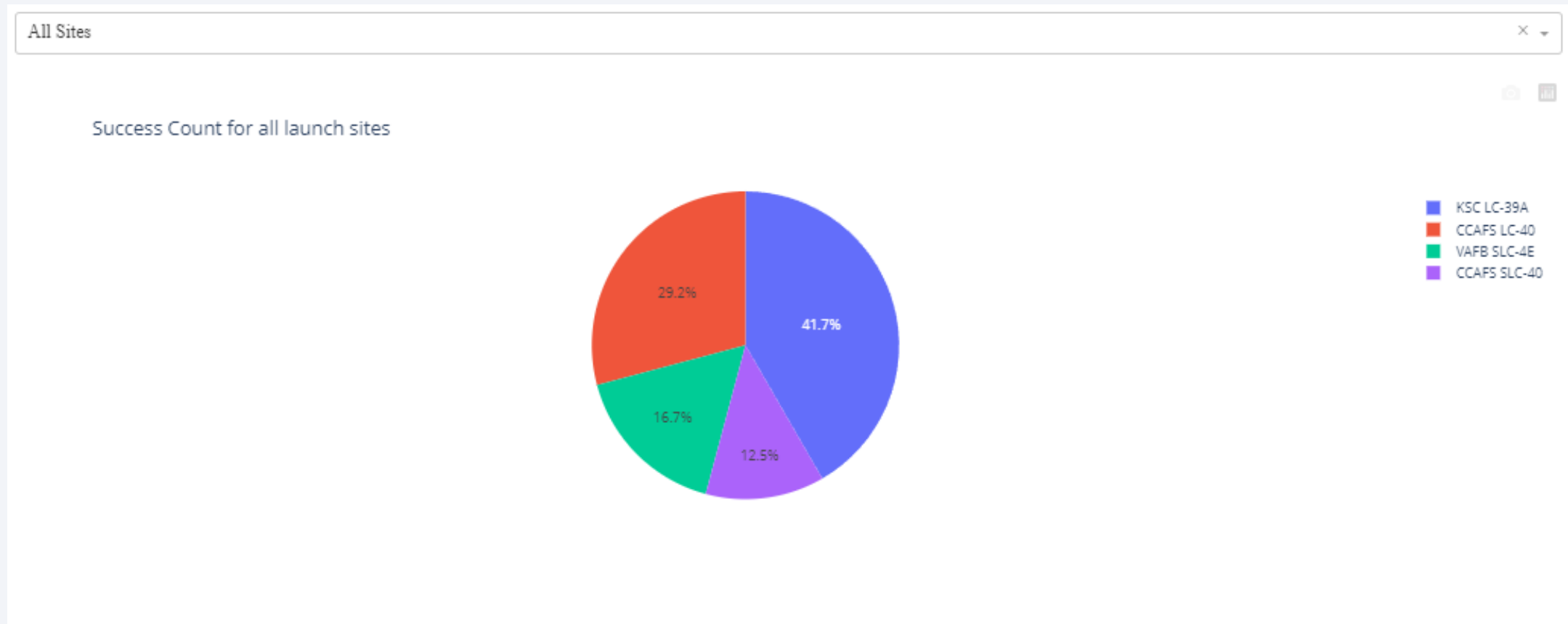


Section 4

Build a Dashboard with Plotly Dash

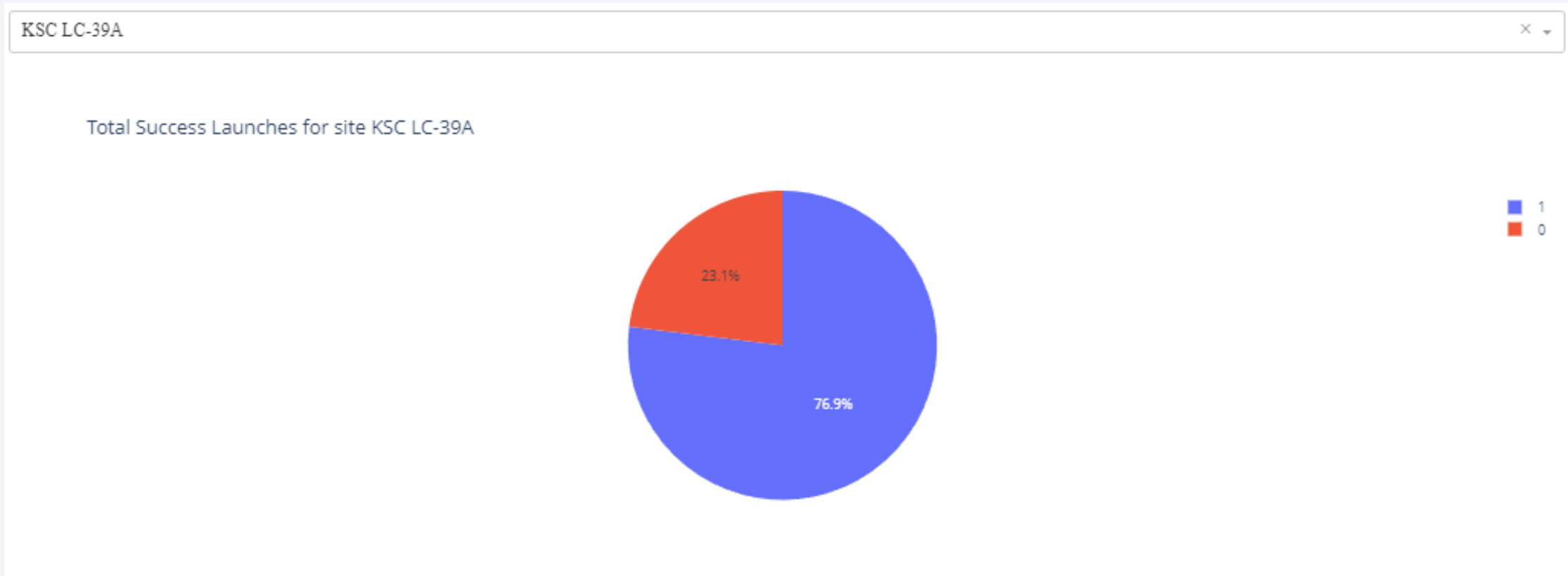
Successful Launches by Site

- The pie chart below shows that KSC LC 39 A has the highest rate of successful launches



Closer look at KSC LC 39 A

- Launch Site KSC LC 39 A has a success rate of about 77% which represents about 3 successful launches for every 1 failed launch



Payload and Launch Outcome



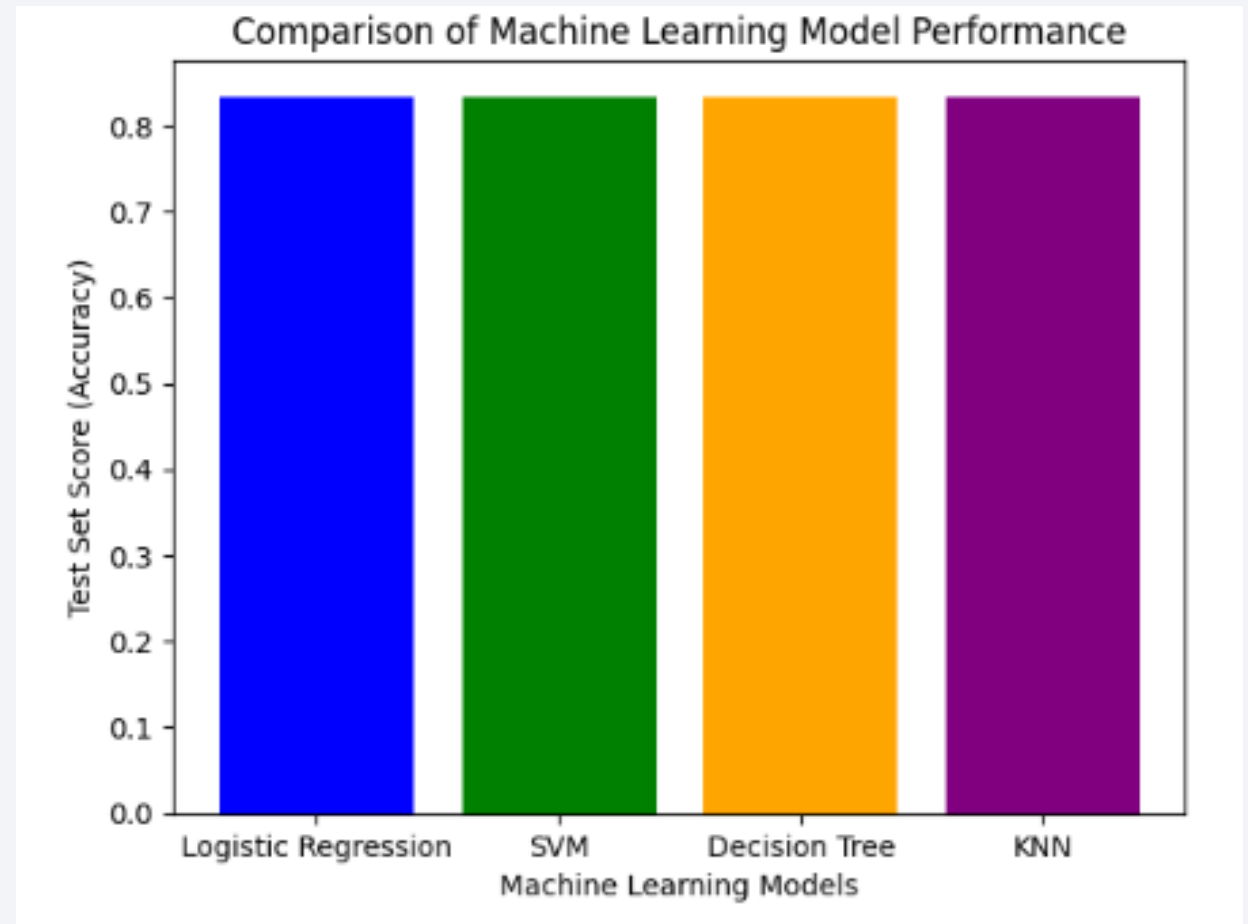
- Top Left: Payload Mass 0 – 5,000 kg
- Top Right: Payload Mass 5,000 – 10,000 kg
- Bottom Left: All Payload Masses 0 – 10,000 kg

Section 5

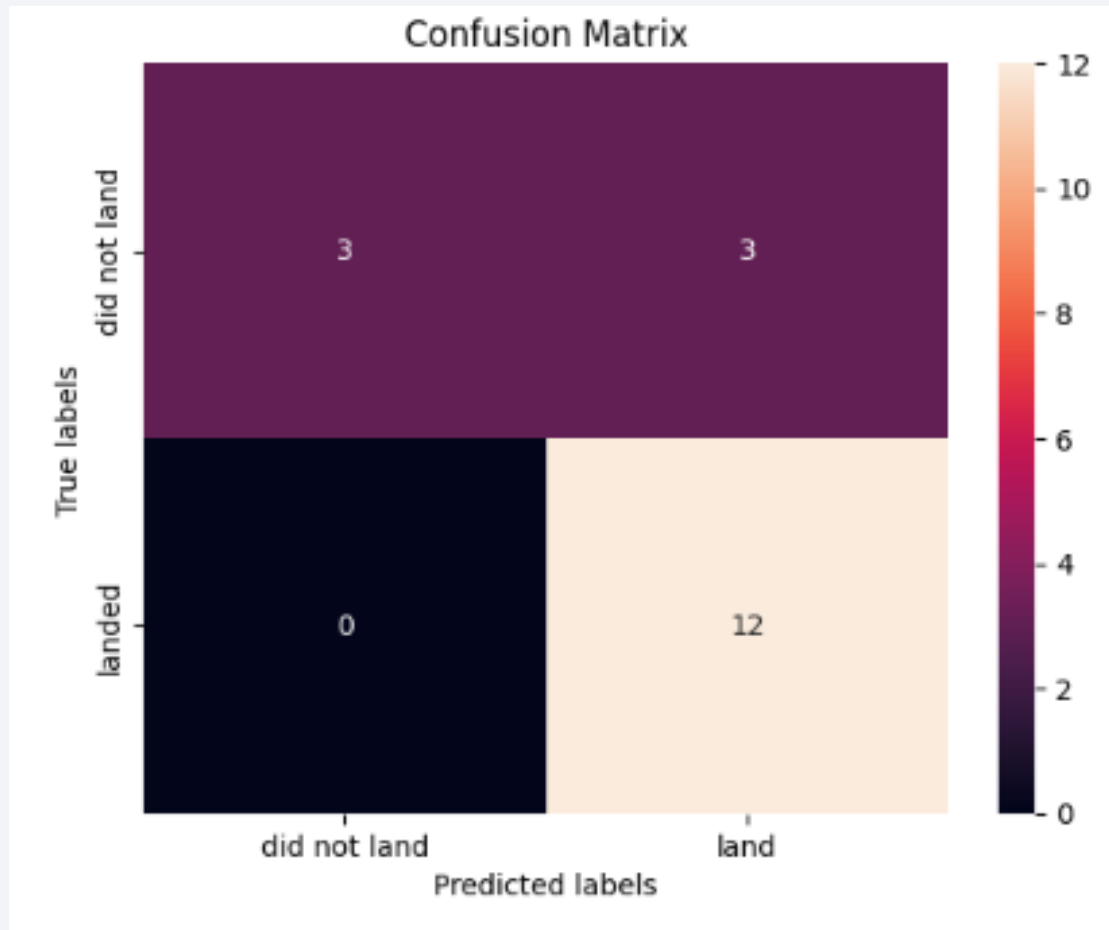
Predictive Analysis (Classification)

Classification Accuracy

- All 4 classification models in testing had the same accuracy score of approximately 83%
- No model outperforms another in this context



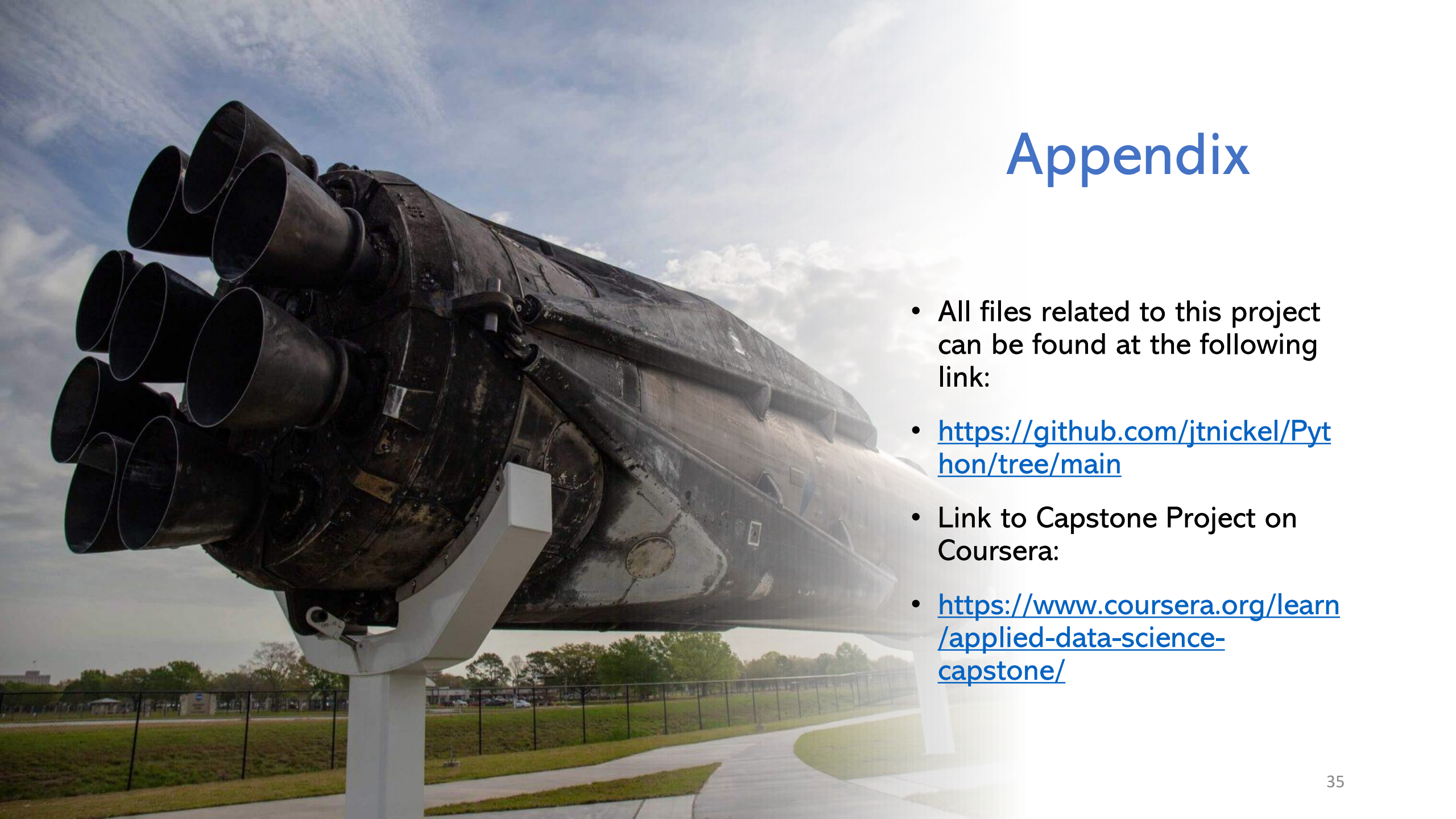
Confusion Matrix



- As all models had the same accuracy score, they also had the same confusion matrix in all 4 cases
- The classifiers give a high number of false positives in each case

Conclusions

- Launch Sites with the most launches tend to have the highest success rate
- LEO, PO, and ISS orbits have the highest success rates for heavier payloads
- Launch Sites are always distanced from high population areas and are always positioned near a coastline for easy retrieval of rockets and safety reasons
- Launch Site KDC LC 39 a has the highest success rate of any site
- All classification models yielded a similar level of accuracy, so any one of them is appropriate for predicting if SpaceX will be able to reuse the first stage on a Falcon 9 rocket launch or not



Appendix

- All files related to this project can be found at the following link:
- <https://github.com/jtnickel/Python/tree/main>
- Link to Capstone Project on Coursera:
- <https://www.coursera.org/learn/applied-data-science-capstone/>

Thank you!

